

## CSE331: Automata and Computability

Assignment 2 | Total: 80 Marks

Deadline: 8th January, 2026

Submission link: <https://forms.gle/CXBZ4VvSTfZ4srqm7>

### Question 1 [10 marks]

Use the pumping lemma and prove the following languages are not regular

- $L1 = \{w \in \{0, 1\}^* : 0^x 1^{y+10} z \text{ where } x = y \text{ and } x, y, z \geq 0\}$
- $L2 = \{w \in \{0, 1\}^* : ww^R, \text{ where } w \text{ is a string and } R \text{ denoting reversed string}\}$
- $L3 = \{w \text{ is not a palindrome}\}, \text{ where } \Sigma = \{0, 1\}$

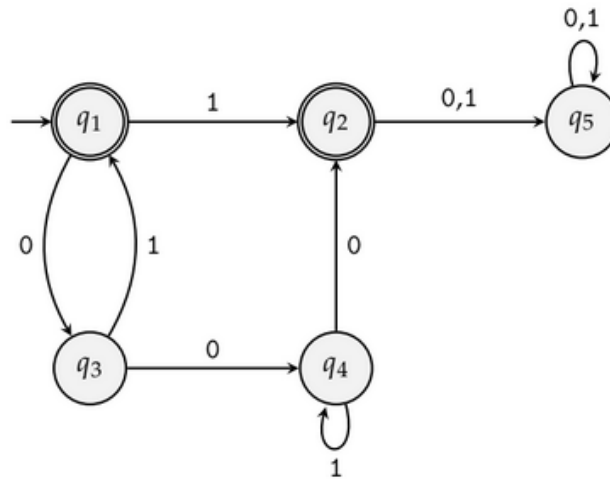
### Problem 2 [10 marks]

Write a CFG for the following CFL:

- $L1 \rightarrow \{0^n 1^m 2^m 3^n \mid n, m > 0\}, \text{ where } \Sigma = \{0, 1, 2, 3\}$
- $L2 \rightarrow \{w = 0^i 1^j 2^k \mid i, j, k \geq 0 \text{ and } j < i + k\}, \text{ where } \Sigma = \{0, 1, 2\}$
- $L3 \rightarrow \{w1\#w2 \mid \text{the number of } 00 \text{ in } w1 \text{ is equal to the number of } 11 \text{ in } w2\},$   
Where  $\Sigma = \{0, 1\}$

### Problem 3 [10 marks]

Convert the following DFA to its corresponding Context Free Grammar.



### Problem 4 [Marks 10]

Take a look at the grammar below and solve the following problems.

$$S \rightarrow aaS \mid abS \mid baS \mid bbS \mid X$$

$$X \rightarrow aaY \mid baY$$

$$Y \rightarrow aY \mid bY \mid \varepsilon$$

- Give a leftmost derivation for the string **abbabbaa**. (3 points)
- Sketch the parse tree corresponding to the derivation you gave in (a). (2 points)
- Demonstrate that the given grammar is ambiguous by showing one more parse tree (apart from the one you already found in (b)) for the same string. (4 points)
- Find a string  $w$  of length nine such that  $w$  has exactly one parse tree in the grammar above. (1 point)

**Problem 5 [10 marks]**

Let  $\Sigma = \{0, 1\}$ . Consider the following pair of languages over  $\Sigma$ .

$$L_1 = \{w \mid \text{the length of } w \text{ is divisible by four}\}$$

$$L_2 = \{w \mid w = 0^{n+2}1^{2n}, n \geq 0\}$$

Now solve the following problems.

- (a) **Construct** a pushdown automaton that recognizes  $L_1$ . (4 points)
- (b) **Construct** a pushdown automaton that recognizes  $L_2$ . (6 points)

**Problem 6 [10 marks]**

For  $\Sigma = \{\#, a, b\}$ , design a Turing machine to recognize the language

- I.  $L = \{a^{2n}\#b^n \mid n \geq 0\}$
- II.  $L = \{ww^R \mid n \geq 0\}$

**Problem 7 [10 marks]**

Prove that the following languages are decidable.

- I.  $L = \{\langle M, w \rangle \mid \text{The Turing machine } M \text{ halts on input } w \text{ within } N \text{ steps}\}$
- II.  $L = \{\langle G \rangle \mid G \text{ is a connected undirected graph}\}$

**Problem 8 [10 marks]**

Show that the following statements are true:

- I. The collection of decidable languages is closed under Union.
- II. The collection of decidable languages is closed under Complement.